

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA0307	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	S.Muskan	Reg. No.:	192472182
Section / Batch:		Date:	

Code of Conduct

I, **S.Muskan (192472182)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **S.Muskan**

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.
Apply rules to decompose each word into root and suffix components.
Differentiate between inflectional and derivational forms within the dataset.
Normalize all variants to a common base form for efficient retrieval.
Determine the parsed structure and normalized output for each word.
2. Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.
Identify prefixes, suffixes, and base forms using rule-based decomposition.
Ensure semantic consistency across derived word forms.
Classify each transformation as inflectional or derivational.
Produce the root and morphological breakdown for each word.
3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.
Apply morphological rules to strip affixes and identify the base form.
Handle both grammatical variations and word formation changes.
Ensure consistent root extraction across all forms.
Determine the stem and classification for each input word.
4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.
Apply state transitions to represent suffix transformations and irregular forms in verb inflection.
Differentiate between regular and irregular inflectional patterns within the dataset.
Ensure consistent mapping of all forms to a common root representation.
Determine the state transitions, morphological breakdown, and normalized output for each word.
5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.
Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.
Handle variations in suffix patterns while preserving semantic consistency.
Ensure uniform root extraction across all derived forms.
Determine the stemming steps, intermediate forms, and final stem for each input word.

Q1. Morphological Analysis Pipeline for Document Indexing

Problem Understanding

A document indexing system should identify the root word and separate affixes from different word forms. This helps store all related words under a single normalized form, improving search efficiency.

Step 1: Morphological Decomposition

Input Word	Root	Affix	Morphological Structure
connected	connect	-ed	connect + ed
connecting	connect	-ing	connect + ing
connection	connect	-ion	connect + ion

Step 2: Classification

Word	Type	Reason
connected	Inflectional	'-ed' indicates past tense without changing word class.
connecting	Inflectional	'-ing' indicates present participle without changing meaning significantly.
connection	Derivational	'-ion' changes the verb into a noun.

Step 3: Normalization

All words are normalized to the common base form:

connect

Morphological Analysis Pipeline

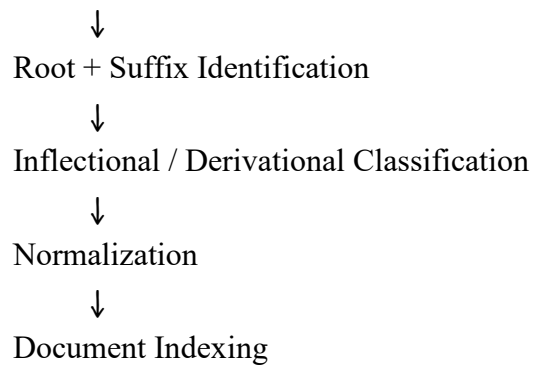
Input Words



Tokenization



Morphological Analysis



Final Output

Word	Normalized Form
connected	connect
connecting	connect
connection	connect

Conclusion

The pipeline successfully extracts the common root **connect**, classifies suffixes correctly, and stores all variants under one indexed representation for efficient information retrieval.

Q2. Morphological Parsing for Sentiment Analysis

Problem Understanding

Morphological parsing separates prefixes and suffixes to identify the base word while preserving semantic meaning for sentiment analysis.

Step 1: Morphological Breakdown

Word	Prefix	Base Word	Suffix
unhappy	un-	happy	—
happiness	—	happy	-ness
happily	—	happy	-ly

Step 2: Classification

Word	Type	Explanation
unhappy	Derivational	Prefix un- reverses the meaning.
happiness	Derivational	-ness converts adjective into noun.
happily	Derivational	-ly converts adjective into adverb.

Semantic Consistency

Although the grammatical forms change, every word originates from the base word:

happy

Morphological Parsing Diagram

happy

- un + happy = unhappy
- happy + ness = happiness
- happy + ly = happily

Final Output

Word	Root
unhappy	happy
happiness	happy
happily	happy

Conclusion

The parser correctly identifies the root **happy**, recognizes prefixes and suffixes, and preserves semantic relationships among all derived forms.

Q3. Stemming-Based Preprocessing Module

Problem Understanding

A search engine should reduce different forms of the same word to a common stem so that a query retrieves all relevant documents.

Step 1: Morphological Analysis

Word	Root	Affix
played	play	-ed
player	play	-er
playing	play	-ing

Step 2: Stemming

Word	Stem	Type
played	play	Inflectional
player	play	Derivational
playing	play	Inflectional

Stemming Process

played

↓ *Remove -ed*

play

player

↓ *Remove -er*

play

playing

↓ *Remove -ing*

play

Final Output

Common Stem:

play

Conclusion

All grammatical and derived forms are reduced to **play**, improving indexing and search accuracy.

Q4. Finite-State Morphological Parsing

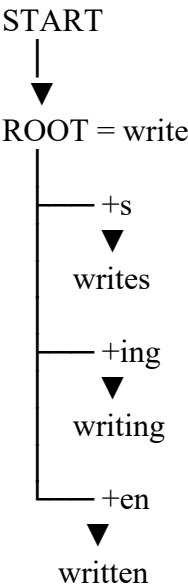
Problem Understanding

Finite-State Morphological Analysis uses states and transitions to generate or recognize word forms while mapping them back to a common root.

Step 1: Morphological Breakdown

Word	Root	Affix	Pattern
writes	write	-s	Regular
writing	write	-ing	Regular
written	write	-en	Irregular

Finite-State Transitions



Classification

Word	Type
writes	Regular Inflection
writing	Regular Inflection
written	Irregular Inflection

Normalized Form

All words map to: **write**

Conclusion: The finite-state parser successfully handles both regular and irregular verb forms and normalizes them to the root **write**.

Q5. Porter Stemmer-Based Processing

Problem Understanding

The Porter Stemmer removes derivational suffixes in multiple stages to obtain a common stem used for document retrieval.

Step-by-Step Stemming

Word 1: relational

relational

↓ *Remove -ational*

relation

↓ *Remove -ion*

relat

Word 2: relation

relation

↓ *Remove -ion*

relat

Word 3: relate

relate

↓ *Remove final -e*

relat

Intermediate Forms

Word	Intermediate Form	Final Stem
relational	relation	relat
relation	relation	relat
relate	relate	relat

Final Output

Common Stem:

relat

Conclusion

The Porter Stemmer removes derivational suffixes while preserving the semantic root. All three words are reduced to the common stem **relat**, ensuring consistent indexing and document retrieval.

Rubrics for Calculation

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____